Solid State Physics — Week 10 Assignment Questions & Answers

Question 91

The conductivity of the semiconductor follows the given equation $\left(\sigma = Ae^{\frac{-E_g}{2K_BT}}\right)$, where A is constant and independent of temperature and K_B is the Boltzmann constant. The conductivity of Ge is $0.7\ \Omega^{-1}m^{-1}$ at 0°C and $2\ \Omega^{-1}m^{-1}$ at 20°C. what is the average band gap ofGe? (Boltzmann constant$K_B = 1.38 \times 10^{-23}\ J/K$)

(a) $0.72\ eV$

(b) $0.65\ eV$

(c) $0.55\ eV$

(d) $0.43\ eV$

Accepted answer: (a)

Question 92

Find out the conductivity of pure silicon at room temperature (300K). Given: mobility of electron$\left(\mu_n = 0.13\frac{m^2}{Vs}\right)$, mobility of hole $\left(\mu_h = 0.05\frac{m^2}{Vs}\right)$ and intrinsic charge carrier concentration$(n_i = 1.5 \times 10^{16}/m^3)$.

(a) $3.12 \times 10^{-3}\ \Omega^{-1}m^{-1}$

(b) $4.32 \times 10^{-4}\ \Omega^{-1}m^{-1}$

(c) $5.12 \times 10^{-5}\ \Omega^{-1}m^{-1}$

(d) $6.64 \times 10^{-6}\ \Omega^{-1}m^{-1}$

Accepted answer: (b)

Question 93

Calculate the resistance of a bar of pure silicon of length 1 cm and cross-sectional area $1mm^2$ at 300 K. Given: mobility of electron$\left(\mu_n = 0.14\frac{m^2}{Vs}\right)$, mobility of hole $\left(\mu_h = 0.07\frac{m^2}{Vs}\right)$ and intrinsic charge carrier concentration$(n_i = 2 \times 10^{16}/m^3)$.

(a) 12.34 $M\Omega$

(b) 14.88 $M\Omega$

(c) 23.15 $M\Omega$

(d) 46.28 $M\Omega$

Accepted answer: (b)

Question 94

Find the conductivity of p-type germanium crystal which is doped with acceptor atoms of concentration $2 \times 10^{23}\ atoms/m^3$ and all the acceptor atoms are active. Mobility of hole in the crystal is$\left(\mu_h = 0.19\frac{m^2}{Vs}\right)$.

(a) 6080 $\Omega^{-1}m^{-1}$

(b) 4038 $\Omega^{-1}m^{-1}$

(c) 3125 $\Omega^{-1}m^{-1}$

(d) 2215 $\Omega^{-1}m^{-1}$

Accepted answer: (a)

Question 95

Calculate the concentration of acceptor atoms that have to be added to an intrinsic semiconductor to produce a p-type semiconductor of conductivity$\left(\sigma_h = 0.05\ \Omega^{-1}m^{-1}\right)$. The mobility of holes in p-type semiconductor is$\left(\mu_h = 0.4\frac{m^2}{Vs}\right)$. Consider all the acceptor atoms are active.

(a) $2.5 \times 10^{28}/m^3$

(b) $1.7 \times 10^{23}/m^3$

(c) $7.8 \times 10^{17}/m^3$

(d) $4.6 \times 10^{15}/m^3$

Accepted answer: (c)

Question 96

The energy gap in Ge is$0.75\ eV$. Compare the intrinsic conductivity of Ge at 27°C and 57°C. Assuming the conductivity of intrinsic semiconductor follows by the given equation$\left(\sigma = CT^{\frac{3}{2}}e^{\frac{-E_g}{2K_BT}}\right)$. Here C is the constant and Boltzmann constant$(K_B = 1.38 \times 10^{-23}\ J/K)$.

(a) 0.23

(b) 0.48

(c) 0.62

(d) 0.78

Accepted answer: (a)

Question 97

Silicon is doped with boron to a concentration of$(4 \times 10^{26}/m^3)$. Assuming the intrinsic concentration of silicon is$(n_i = 1.5 \times 10^{16}/m^3)$. Find the position of Fermi level of doped semiconductor with respect to the intrinsic Fermi level at room temperature 300 K. Boltzmann constant$(K_B = 1.38 \times 10^{-23}\ J/K)$.

(a) 0.62 eV below to the intrinsic Fermi level

(b) 0.44 eV below to the intrinsic Fermi level

(c) 0.32 eV below to the intrinsic Fermi level

(d) 0.27 eV below to the intrinsic Fermi level

Accepted answer: (b)

Question 98

A sample of Ge is doped to the extent of$(10^{20}\ donor\ atoms/m^3)$ and$(5 \times 10^{19}\ acceptor\ atoms/m^3)$. At 300 K, the resistivity of intrinsic Ge is$(0.6\ \Omega - m)$. If the applied electric field is 200 V/m, find the total conduction current density. Assume the ratio of hole mobility to electron mobility $\frac{\mu_h}{\mu_n} = \frac{1}{2}$ and intrinsic carrier density $(n_i = 2.5 \times 10^{19}/m^3)$ at 300 K. Consider all the donor and acceptor atoms are active in Ge.

(a) 0.058 $\frac{A}{cm^2}$

(b) 0.078 $\frac{A}{cm^2}$

(c) 0.098 $\frac{A}{cm^2}$

(d) 1.048 $\frac{A}{cm^2}$

Accepted answer: (a)

Question 99

In an intrinsic semiconductor, the effective mass of the electron is $0.09m_e$ and that of the hole is$(0.4m_e)$ where (m_e) the electrons rest mass. Calculate the intrinsic carrier concentration at 300K, when the energy gap$(E_g = 0.7\ eV)$. Consider Boltzmann constant$(K_B = 1.38 \times 10^{-23}\ J/K)$ and Planck constant is$(h = 6.62 \times 10^{-34} J - s)$.

(a) $(2.7 \times 10^{18}\ /m^3)$

(b) $(3.2 \times 10^{20}\ /m^3)$

(c) $(4.5 \times 10^{28}\ /m^3)$

(d) $(6.2 \times 10^{32}\ /m^3)$

Accepted answer: (a)

Question 100

At a given temperature, a semiconductor with intrinsic carrier concentration $(n_i = 10^{16}/m^3)$ is doped with a donor dopant of concentration$(n_D = 10^{26}/m^3)$. If the temperature remains the same, calculate the concentration of hole in the doped semiconductor.

(a) $10^4/m^3$

(b) $10^6/m^3$

(c) $10^8/m^3$

(d) $10^{10}/m^3$

Accepted answer: (b)